Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Weight and Centre of Mass

1 · Mass and weight

The rule

Mass is the amount of matter in an object. It is measured in **kilograms (kg)** and is the same everywhere.

Weight is the force on an object caused by gravity. It is measured in **newtons (N)** with a newton-meter, and it depends on where the object is.

$$W = m\,g$$

- W → **weight** (N)
- m → **mass** (kg)
- g → **gravitational field strength** (N/kg)

To find the weight: $W = m \times g$

To find the mass: $m = W \div g$

To find the field strength: $g = W \div m$

On Earth, g = **9.8 N/kg**: every 1 kg of mass has a weight of 9.8 N.

Mass goes in kilograms. Divide grams by 1000: 250 g $= 250 \div 1000 = 0.25$ kg.

Worked example — copy this shape

Find the weight. A dog has a mass of 15 kg. Calculate its weight on Earth.

Equation $W = m \quad g$

Substitute $W = 15 \times 9.8$

Calculate $W =$ **147 N**

Find the mass. A rover weighs 4000 N on Mars, where $g = 3.7$ N/kg. Calculate its mass.

Equation $W = m \quad g$

Substitute $4000 = m \times 3.7$

Rearrange $m = 4000 \div 3.7$

Calculate $m =$ **1081 kg**

Check it. Put your answer back into $W = m \times g$: $15 \times 9.8 = 147$ N, the weight you found. A mass in kg and a weight in N are different quantities, so each answer carries its own unit.

Task 1 · Use $W = mg$

Fill in each gap. Use the shape of the worked example and give every answer with its unit. Use $g = 9.8$ N/kg on Earth. Rows 6 and 7 start in grams.

#	Object	Mass	g	Weight
1	a school bag, on Earth	6 kg	9.8 N/kg	
2	Jenny, on Earth	65 kg	9.8 N/kg	
3	a Great Dane, on the Moon	40 kg	1.6 N/kg	
4	a Great Dane, on Jupiter	40 kg	25 N/kg	
5	a satellite, on Earth		9.8 N/kg	4900 N
6	a caterpillar, on Earth	20 g	9.8 N/kg	
7	a phone, on Earth	150 g	9.8 N/kg	
8	the Martian, on Mars	120 kg		444 N

The same Great Dane has a mass of _____ on the Moon, on Jupiter and on Earth. Its _____ is the quantity that changes, because g changes.

2 · Practical: measure mass and weight

Method. Collect four masses. For each one, (1) read its mass from the balance in grams and convert to kilograms, (2) hang it from the newton-meter and read the weight in newtons, (3) work out weight ÷ mass.

Mass / g	Mass / kg	Weight / N	Weight ÷ mass / N/kg

Conclusion. Weight ÷ mass comes out at about _____ N/kg every time. That is the value of g on Earth. On Earth, **100 g** has a weight of about _____.

3 · Centre of mass

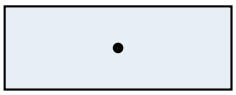
The rule

The weight of an object acts as if it were a single force at one point: the **centre of mass**. Draw the weight arrow **from the centre of mass**, straight down, and label it **weight**.

A hanging object comes to rest with its centre of mass directly below the point it hangs from.

Task 2 · Draw the weight

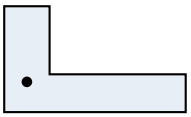
The centre of mass of each object is marked with a dot. Draw the weight arrow from the dot, about 1 cm long, and label it **weight**.



a metre-rule's block



a football



an L-shaped bracket

4 · Grams, kilograms, newtons

Tomorrow, in the force and acceleration practical, masses hang from a string over a pulley. The **weight of the hanging masses** is the force that accelerates the trolley. Each mass is stamped in grams, so the order is always the same:

grams $\xrightarrow{\div 1000}$ kilograms $\xrightarrow{\times 9.8}$ newtons

Worked example — copy this shape

A 50 g mass hangs on the string. Calculate the force it applies.

Grams to kg $m = 50 \div 1000 = 0.05$ kg

Equation $W = m \quad g$

Substitute $W = 0.05 \times 9.8$

Calculate $W =$ **0.49 N**

Task 3 · Weight of the hanging masses

(a) A mass hanger and masses together have a mass of 100 g. Calculate their weight.

[2]

(b) Masses of 20 g, 20 g and 10 g are added to a 50 g hanger. Calculate the total weight now pulling on the string.

[3]

(c) The force on the trolley is to be 0.49 N. Calculate the total mass that must hang, in grams.

[3]

Check it. Every 100 g you add makes the weight about 1 N bigger; your answers to (a) and (b) agree with that.